

Middle East respiratory syndrome coronavirus - Global update

24 December 2025

Situation at a glance

Since the beginning of 2025 and as of 21 December 2025, a total of 19 cases of Middle East respiratory syndrome coronavirus (MERS-CoV), including four deaths have been reported to WHO globally. Of the 19 cases, 17 were reported by the Kingdom of Saudi Arabia (KSA), and two were reported from France. Between 4 June and 21 December 2025, the Ministry of Health (MoH) of KSA reported a total of seven cases of MERS-CoV infection, including two deaths. In addition, at the beginning of December 2025, the National IHR Focal Point (IHR NFP) for France also reported two MERS-CoV travel – associated cases; involving individuals with recent travel to countries in the Arabian Peninsula. The notification of these latest cases does not change the overall risk assessment, which remains moderate at both the global and regional levels. These cases show that the virus continues to pose a threat in countries where it is circulating in dromedary camels, with regular spillover into the human population. WHO recommends implementation of targeted infection, prevention and control (IPC) measures to prevent the spread of health care-associated infections of MERS-CoV and onward human transmission.

Description of the situation

Since the first report of MERS-CoV in the KSA and Jordan in 2012, a total 2635 laboratory-confirmed cases of MERS-CoV infection, with 964 associated deaths (Case Fatality Ratio (CFR) of 37%), have been reported to WHO from 27 countries, across all six WHO regions (Figure 1). The majority of cases (84%; n=2224), have been reported from the KSA (Figure 2). Since the beginning of 2025 and as of 21 December, a total of 19 cases have been reported to WHO. Overall, 17 cases were reported in the KSA from five regions named: Riyadh (n=10), Taif (n=3), Najran (n=2), Hail (n=1), and Hafr Al-Batin City (n=1) (Figure 3). In addition, two travel associated cases of MERS-CoV infection have been reported in France, with likely exposure occurring during recent travel in the Arabian Peninsula (Figure 3).

This disease outbreak news report focuses on the recent nine cases of MERS-CoV infection reported between 4 June - 21 December 2025: seven cases from the KSA and the two imported cases to France. The details of cases reported earlier in 2025 can be referred to in the previously published disease outbreak news on [13 March 2025](#) and [12 May 2025](#).

Between 4 June and 21 December 2025, the MoH of the KSA reported a total of seven cases of MERS CoV infection. The cases were reported from three regions: Najran (2), Riyadh (3), and Taif (2). No epidemiological links were identified between the seven cases. In addition, between 2 and 3 of December 2025, the IHR NFP for France reported two cases of MERS – CoV with recent travel to the Arabian Peninsula during the month of November.

Follow-up has been completed for all contacts and no secondary infections have been identified or reported. From September 2012, France has recorded a total of four laboratory-confirmed cases of MERS-CoV infection, including one death: two cases were reported in 2013, and the latest two cases in December 2025. All cases had been travelers exposed in the Arabian Peninsula and returning back to France.

For additional details please see Table 1.

Figure 1: Epidemic curve of MERS-CoV infections (2635) and deaths (964) reported

globally between 2012-2025

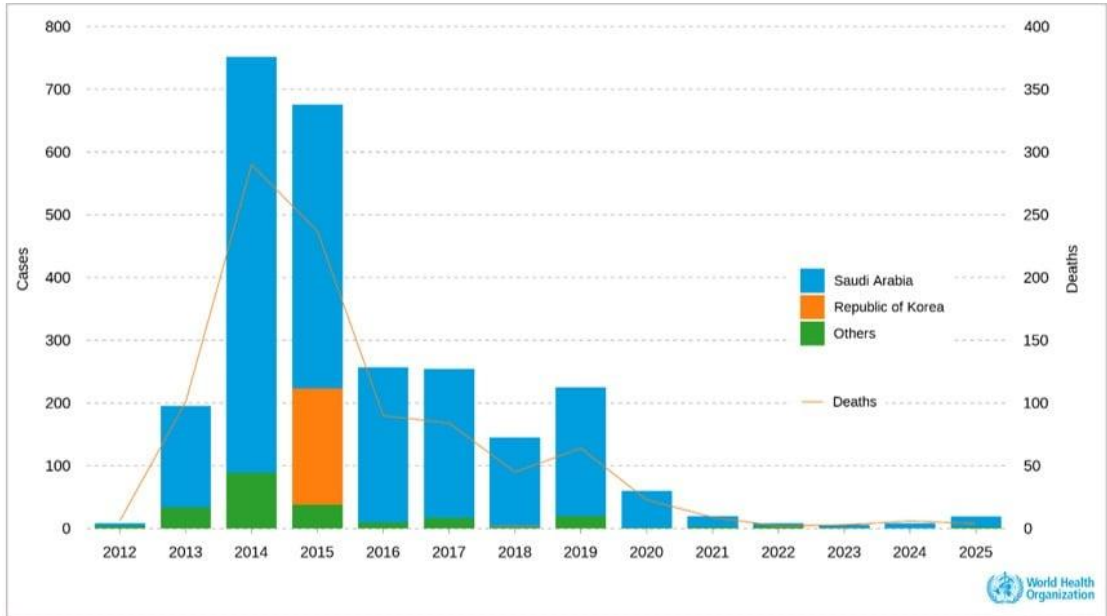


Figure 2: Epidemic curve of MERS-CoV infections (2224) and deaths (868) reported in KSA between 2012-2025

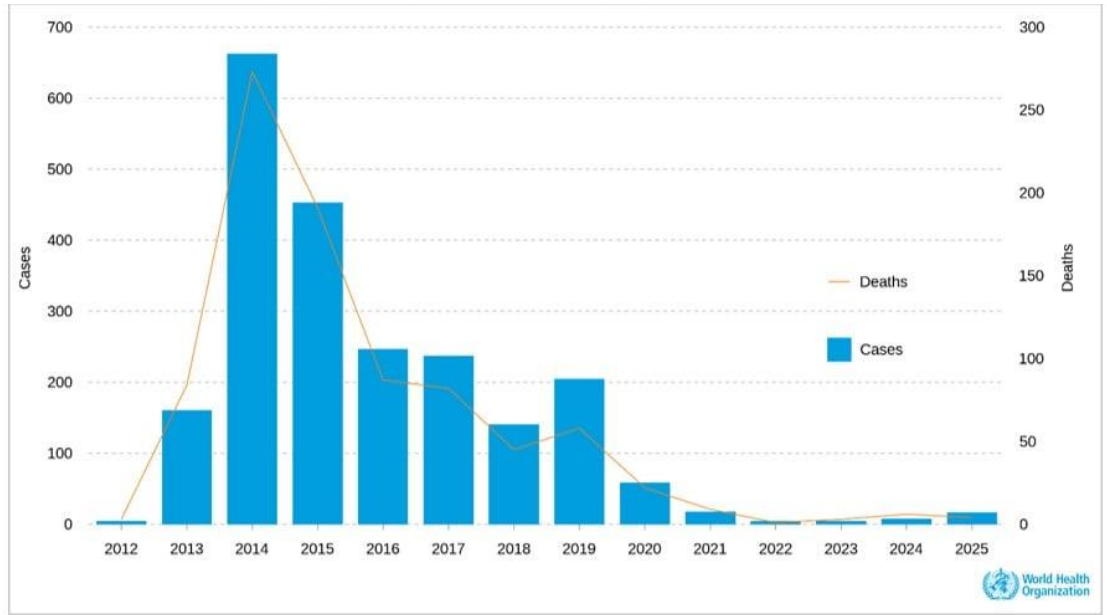


Figure 3. Geographical distribution of MERS-CoV infections between 1 January and 21 December 2025 (n=19).

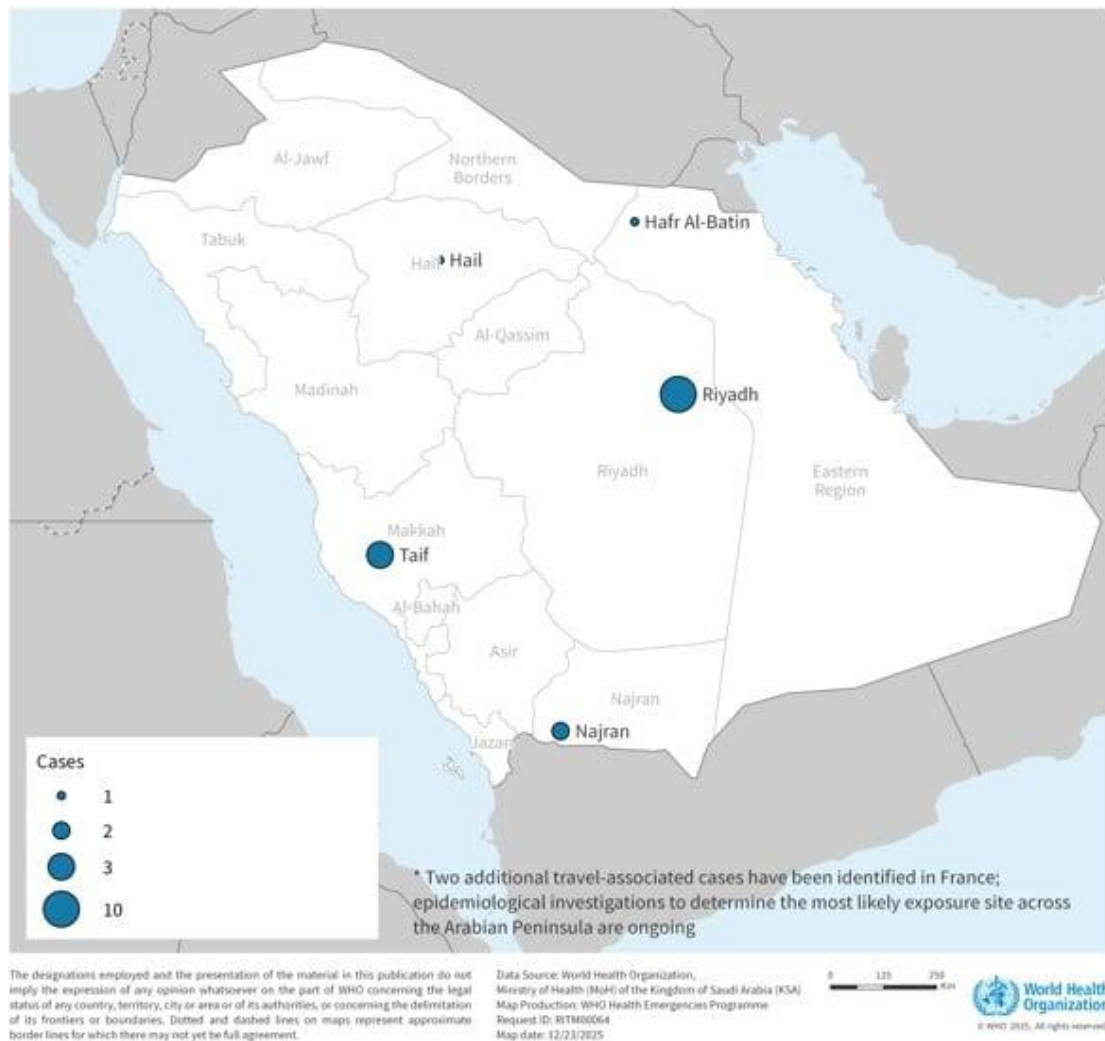


Table 1: MERS-CoV cases reported by KSA and France between 4 June and 21 December 2025

Case number	Region, Country of residence	Age in years	Health and care worker	Exposure	Date of symptom onset	Symptoms	Date of first hospitalization	Date of sample collected	Date of laboratory confirmation	Type of Laboratory test	Comorbidities	Health status at the time of notification
Case 1	Taif, KSA	18-65	No	Indirect contact with camels	25 May	Fever, cough, shortness of breath	30 May	1 June	4 June	RT-PCR	Yes	Recovered
Case 2	Riyadh, KSA	18-65	No	No known history of contact with camels or camel products	29 July	Fever, shivering	29 July	11 August	12 August	RT-PCR	Yes	Deceased
Case 3	Riyadh, KSA	>65	No	No known history of contact with camels or camel products	21 September	Fever, shortness of breath	26 September	28 September	1 October	RT-PCR	No	Deceased
Case 4	Najran, KSA	18-65	No	No known history of contact with camels or camel products	20 October	Shortness of breath	14 October (already admitted due to unrelated condition)	27 October	27 October	RT-PCR	Yes	Recovered
Case 5	Taif, KSA	>65	No	No known history of contact with camels or camel products	16 November	Cough, shortness of breath	16 November	18 November	19 November	RT-PCR	Yes	Recovered
Case 6	Riyadh, KSA	>65	No	Direct contact with camels	26 November	Shortness of breath, tachypnea	24 November (already admitted due to unrelated condition)	30 November	30 November	RT-PCR	Yes	Still under treatment
Case 7	Najran, KSA	18-65	No	No known history of contact with camels	10 December	Fever, cough, and shortness of breath	13 December	15 December	16 December	RT-PCR	Yes	Still under treatment
Case 8	France	>65	No	Consumption of camel products (meat) and dates + animal market visits	24 November	Cough, fever, dyspnea	26 November	28 November	29 November	RT-PCR	Yes	Still hospitalized for monitoring. The patient is doing well.
Case 9	France	>65	No	Consumption of camel products (meat) and dates + animal market visits	23 November	Rhinitis	3 December	1 December	2 December	RT-PCR	No	Recovered. Discharged on 10/12/2025

Epidemiology

Middle East respiratory syndrome (MERS) is a respiratory illness caused by a coronavirus (MERS-CoV). The case fatality ratio (CFR) among confirmed cases is around 37%. The CFR is calculated based solely on laboratory-confirmed infections and may overestimate the actual mortality rate since milder cases often go undetected or unreported.

Humans can contract MERS-CoV through multiple transmission pathways; the primary route being through direct or indirect contact with dromedary camels, which serve as the virus's natural host and primary zoonotic reservoir. Additionally, human-to-human transmission can occur via infectious respiratory particles primarily in close-contact situations and can also occur through direct or indirect contact; this is especially prominent in health-care settings. Human-to-human transmission of the virus has occurred in health care facilities in several countries, including transmission from patients to health care providers and transmission between patients before MERS-CoV was diagnosed. It is not always possible to identify patients with MERS-CoV early or without testing because symptoms and other clinical features may be non-specific. Outside these environments, there has been limited documented human-to-human transmission.

MERS can present with no symptoms (asymptomatic), mild symptoms (including mild respiratory issues), or severe illness leading to acute respiratory distress and death. Common symptoms include fever, cough, and breathing difficulties, with pneumonia frequently observed, though not always present. Some patients also experience gastrointestinal symptoms such as diarrhoea. Severe cases may require intensive care, including mechanical ventilation. Those at higher risk of severe outcomes include older adults, individuals with weakened immune systems, and those with chronic conditions like diabetes, kidney disease, cancer, or lung disorders.

The number of MERS-CoV infections reported to WHO substantially declined since the beginning of the COVID-19 pandemic. Initially, this was likely the result of epidemiological surveillance for SARS-CoV-2 being prioritized. Similar clinical pictures of both diseases may have resulted in reduced testing and detection of MERS-CoV infections. However, the MoH of the KSA has been working to improve testing capacities for better detection of MERS-CoV since the easing of the COVID-19 pandemic, with MERS-CoV included into sentinel surveillance testing algorithms since the second quarter of 2023, for samples that test negative for both influenza and SARS-CoV-2. In addition, recommended IPC measures (e.g., mask-wearing, hand hygiene, physical distancing, improving ventilation) and public health and social measures in the community to reduce SARS-CoV-2 transmission, (stay-at-home orders, reduced mobility) also likely reduced onward human-to-human transmission of respiratory infections including MERS-CoV. Potential cross-protection conferred from infection with or vaccination against SARS-CoV-2 and any reduction in MERS-CoV infection or disease severity and vice versa has been hypothesized but requires further investigation. ^[1,2]

Public health response

WHO is supporting Member States in strengthening preparedness and response.

Activities in the Kingdom of Saudi Arabia include;

- Strengthened surveillance with immediate notification of all suspected and confirmed cases.
- Strict implementation of infection prevention and control transmission-based precautions (Contact and Droplet precautions) in healthcare facilities for suspect or confirmed patients, and airborne precautions for patients undergoing aerosol-generating procedures.
- Identification of health and care worker contacts and perform risk assessment of

their exposure, considering the timely identification of symptomatic patients, implementation of IPC measures, and correct utilization of PPE while treating patients,

- Exposed health and care workers are followed up for 14 days to monitor symptoms. If they develop symptoms, they are to be removed from working with patients until tested and symptoms are fully resolved.
- Patients exposed to MERS-CoV in the healthcare setting must be tested to determine their ability to continue working with patients without further transmission, which could potentially lead to outbreaks in the healthcare facility.
- Identification of all potential community contacts and active follow-up to monitor symptoms for 14 days.
- All community acquired cases are investigated for having direct or indirect contact with camels or their products.
- Cases linked to camel exposures are notified to the National Center for Prevention and Control of Plants, Pests, and Animal Diseases (Weqaa) to investigate potential camel sources.
- Camels identified as a presumed source are quarantined and tested for MERS-CoV, and if live virus is detected, the quarantine period will be extended until live virus is no longer detected in camel.

Activities in France include;

- On 4 December 2025, MoH France published information regarding the two imported cases of MERS-CoV in the country ([link](#)).
- Genomic sequencing was conducted from the first case and reported as being the same lineage that is circulating in the Arabian Peninsula. Further laboratory analyses are ongoing.
- Contact tracing was initiated as soon as the first case was detected for the monitoring and surveillance of fellow travellers and co-exposed individuals, high-risk contacts, and hospital contacts. It was completed in week 51 and no additional cases among the travellers have been reported, nor any secondary cases as of 19 December 2025.
- Asymptomatic co-exposed individuals and at-risk contacts located in France were offered a full testing protocol (nasopharyngeal swab, sputum, rectal swab and serology) on a voluntary basis up to 29 days after their last exposure, even if they did not exhibit any symptoms.